From cohort state transitions to age–calendar transport: a reproducible framework for vaccination programme analysis

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Short title: Age–calendar vaccination models

Mathematical methods with a pneumococcal vaccination application in Viet Nam

Abstract

Understanding and modelling vaccination effectiveness is important for national health policy. Health-economic analyses that lump chronological age and dose history can conceal age- and dose-dependent risks, particularly in young children. We formulate a hybrid age–calendar transport framework combining state transitions, continuing births, scheduled-dose interfaces and population feedback. We connect matched ordinary differential equation (ODE) and continuous-time Markov representations in a controlled verification chain: the ODE computes expected counts, and the Markov law specifies conditional terminal variation and event probabilities. The vaccine-programme application is deterministic. We state the conditions linking these representations, derive characteristic and population balances, and show why dose-state aggregation generally fails to close. A five-state pneumococcal conjugate vaccine (PCV) application provides a monthly implementation and numerical refinement study. A separate closed-cohort simulation checks matched discrete-time expected rewards. Populations with identical age and dose marginals produced different indirect multipliers and disease-event totals. Manufactured-solution and submonthly experiments quantified numerical behaviour. For three separately evaluated products, delayed/incomplete dosing, attained-age waning and gradual indirect-protection build-up jointly reduced quality-adjusted life-year (QALY) gains by 10.2–18.4%; paired herd-off comparisons attributed 61.0–69.7% of full-scenario gains to the modelled indirect pathway. Schedule experiments and deterministic perturbations demonstrate how the retained structure changes programme comparisons. The framework connects model representations and identifies evidence needed for programme evaluation. National policy prediction requires empirical calibration, verified valuation inputs and joint parameter uncertainty.

1 Introduction

Vaccination models translate assumptions about disease, dose receipt and protection into population outcomes. State-transition models make event and economic accounting explicit. Their interpretation depends on which state information is retained. A cohort model follows one group from birth, whereas an ongoing programme simultaneously contains children born in different calendar months and at different stages of vaccination. These settings require different population boundaries and state information.

Continuous-time Markov chains (CTMCs) provide forward equations for state probabilities, and mean-field constructions connect individual transition mechanisms to ordinary differential equations (ODEs) under stated conditions [1]. Age-structured transport provides a related representation in which individuals move deterministically through age while changing programme state. Connections between stochastic individual

models and McKendrick–von Foerster partial differential equations (PDEs) are established [2]. Age-stratified ODEs and sufficiently augmented Markov or individual models can represent the same mechanisms. The reason to use transport here is its transparent treatment of characteristics, birth boundaries, scheduled interventions and age-weighted functionals.

In earlier closed-cohort work underlying this project, we used a Markov representation to investigate pneumococcal conjugate vaccines (PCVs), alongside a deterministic implementation of matched expected outcomes. Those analyses followed age through cohort time but represented vaccination through aggregate protection assumptions, without retaining an explicit joint distribution of programme calendar time, chronological age and dose history. The earlier model comparisons are documented in the accompanying repository. Indirect effects can materially change the health-economic evaluation of vaccination programmes [3, 4]. Trials in Vietnamese infants have examined how schedules affect carriage [5]; surveillance and local effectiveness studies also motivate explicit distinctions between individual protection and population effects [6, 7]. Parameter-specific evidence status is recorded separately in the source ledger.

In this paper, we extend our earlier cohort analyses to a continuously renewed structured population and ask three methodological questions. Under which restrictions are state-transition and cohort representations consistent? Which information is lost when an age–dose distribution is reduced to totals? How can scheduling and indirect-protection assumptions be evaluated without confounding a change in mechanism with a change in implementation? We answer these questions by derivation, constructed counterexamples and numerical verification, then use a three-product PCV application to demonstrate trajectory, factorial and counterfactual analyses. Disease and economic quantities illustrate output functionals under a frozen scenario ledger. Reporting separates mathematical properties, computational verification and empirical evidence [8–10].

2 Methods

2.1 State-transition, cohort and transport representations

Let $p(t)$ be a column vector of state probabilities and $Q(t)$ a row generator with non-negative off-diagonal entries and zero row sums, including death as an absorbing state. For externally specified, bounded transition rates,

$$p'(t) = Q(t)^\top p(t), \qquad p(0) = p_0. \tag{1}$$

For constant Q, $p(t) = \exp(tQ^\top)p_0$. Independent individuals following this process have expected state counts $Np(t)$, with count distributions determined by the underlying Markov law. If transition intensities depend on a finite random population, generally $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$; nonlinear population feedback requires

a closure argument or approximation. For example, if $X = N^{-1}\,\mathrm{Binomial}(N, p)$, then $\mathbb{E}[X^2] = p^2 + p(1-p)/N$. At $N = 100$ and $p = 0.5$, this equals 0.2525, whereas $(\mathbb{E}[X])^2 = 0.25$.

A discrete-time model instead satisfies $p_{n+1} = P_n^\top p_n$. With a constant continuous generator over a cycle of length h, the compatible matrix is $P_n = \exp(hQ_n)$. Arbitrary monthly kernels need not be representable by a single such generator. The application includes scheduled-dose kernels and therefore uses a hybrid formulation rather than converting every visit probability into a constant transition hazard.

Let $A_g(t, a)$ denote alive density in programme state g, at calendar time t and chronological age a. Ages and times are measured in months. The general structured formulation, between scheduled-dose interfaces, is

$$(\partial_t + \partial_a)A(t, a) = K(t, a; H[A])^\top A(t, a), \quad 0 < a < L, \quad A(t, 0) = b(t), \quad A(0, a) = A_0(a). \tag{2}$$

The alive-state matrix K is a subgenerator: its off-diagonal entries are non-negative, and row sums are minus death hazards. An absorbing-death augmentation yields a conservative generator. At a scheduled age a_i, a column-stochastic map $J_i(t)$ implements a dose opportunity:

$$A(t, a_i^+) = J_i(t)A(t, a_i^-). \tag{3}$$

The same map may depend on programme calendar time, permitting a declared campaign. Probability is conditional on remaining in the source dose state at that opportunity. In the PCV implementation, $K = -\operatorname{diag}(\mu_g)$ between opportunities; acute diseases are recurrent event rewards, not mutually exclusive disease-state compartments.

Along $a = t - t_0$, Eq. 2 gives a cohort ODE with the same interface maps and the shared feedback supplied by the whole population. A closed-cohort comparison must match initial conditions, coefficients, impulses, feedback and output accounting. A concentrated birth cohort is naturally represented as a measure; its characteristic relation does not require treating a Dirac mass as an ordinary smooth density. The three representations describe the same externally driven cohort under these matched conditions: probability evolution supplies expected state counts, characteristics follow one birth cohort, and transport assembles cohorts with continuing inflow and shared population feedback. Age-resolved ODEs and augmented Markov models can preserve this structure as well.

For a prescribed feedback trajectory h, let $U_h(t, s; a_0)$ denote the cohort propagator starting at age a_0

and time s, including the ordered dose-interface maps. The characteristic representation is

$$A(t,a) = \begin{cases} U_h(t,0;a-t)A_0(a-t), & a > t, \\ U_h(t,t-a;0)b(t-a), & a < t. \end{cases} \tag{4}$$

One-sided traces are used at interfaces; the equality is understood almost everywhere for integrable densities.

2.2 Positivity, balance and the limits of aggregation

Proposition 1 (Positive propagation and characteristic consistency)**.** *Let $A_0 \in L^1_+(0,L;\mathbb{R}^m)$, $b \in L^1_{\mathrm{loc},+}(\mathbb{R}_+;\mathbb{R}^m)$, and let externally prescribed K be measurable and uniformly bounded on each finite time strip, with non-negative off-diagonal entries and non-positive row sums. For a finite number of fixed age interfaces with bounded non-negative column-stochastic maps, the characteristic mild solution is non-negative. Conditional on a prescribed feedback trajectory, it is the same cohort evolution obtained by integrating the matched ODE and applying the same maps.*

Proof. The fundamental solution of the cooperative linear system is non-negative, and multiplication by each non-negative interface map preserves this property. Restricting the transport equation to a characteristic gives the ODE; applying the ordered maps yields the same propagation under the prescribed feedback. □

Proposition 2 (Population balance)**.** *For a piecewise $W^{1,1}$ solution, or a mild solution admitting integrable incoming, outgoing and interface traces, conservative dose maps and death hazard μ_g, total alive population $N(t) = \sum_g \int_0^L A_g(t,a)\,da$ satisfies for almost every t*

$$N'(t) = \mathbf{1}^\top b(t) - \mathbf{1}^\top A(t,L) - \sum_g \int_0^L \mu_g(t,a;H[A])A_g(t,a)\,da. \tag{5}$$

Proof. Integrate the transport equation and sum over states. Internal transition gains and losses cancel. Interface terms cancel because $\mathbf{1}^\top J_i = \mathbf{1}^\top$. The remaining terms are births, age-out and deaths. □

The integral vector $M(t) = \int_0^L A(t,a)\,da$ obeys

$$M'(t) = b(t) - A(t,L) + \int_0^L K(t,a;H[A])^\top A(t,a)\,da + \sum_i (J_i - I)A(t,a_i^-). \tag{6}$$

The weighted reaction integral and the interface and age-out traces require information beyond M in general. In a boundary-free special case, age-independent coefficients and feedback determined solely by M suffice for a closed reaction equation. Exact Markov aggregation similarly requires lumpability conditions [11].

Proposition 3 (Dose-state totals need not determine observables)**.** *Let $R(A) = \sum_g \int_0^L f_g(a) A_g(a)\, da$ with bounded measurable weights. Suppose two distinct states g_1, g_2 and disjoint age intervals E_1, E_2 of positive length satisfy*

$$\Delta_f = \frac{1}{|E_1|} \int_{E_1} (f_{g_1} - f_{g_2})\, da - \frac{1}{|E_2|} \int_{E_2} (f_{g_1} - f_{g_2})\, da \neq 0. \tag{7}$$

Then there exist non-negative L^1 populations with equal dose-state and age marginals but different reward values. A weighted index differs when its numerator satisfies this condition and its common denominator is positive; feedback differs when its map separates the resulting index values.

Proof. Set $u_i = \mathbf{1}_{E_i}/|E_i|$ and choose $m > 0$. Define $A_{g_1} = mu_1$, $A_{g_2} = mu_2$, $B_{g_1} = mu_2$, $B_{g_2} = mu_1$, with identical non-negative densities in all other states. Each selected state has mass m, and both age marginals equal $m(u_1 + u_2)$ plus the common remaining population. Direct integration gives $R(A) - R(B) = m\Delta_f \neq 0$. For an index with numerator weights $f_g = \rho_g w$, the denominator depends only on the common age marginal, so division by its positive value preserves the difference. Applying a separating feedback map completes the construction. Piecewise-constant application weights permit intervals entirely within the declared age bands. □

For local nonlinear well-posedness, let $X = L^1(0, L; \mathbb{R}^m)$ and assume non-negative bounded weights, fixed integrable birth data, and a bounded, strongly continuous transport evolution with the fixed interface maps. Assume the reaction matrices are uniformly bounded on bounded feedback sets and uniformly locally Lipschitz in feedback. Write $D(A) = \sum_g \int w A_g$ and $N_\rho(A) = \sum_g \rho_g \int w A_g$, with $D_0 = D(A_0) > 0$. On a bounded set with $D(A), D(B) \geq \delta > 0$,

$$\left| \frac{N_\rho(A)}{D(A)} - \frac{N_\rho(B)}{D(B)} \right| \leq \frac{|N_\rho(A) - N_\rho(B)|}{\delta} + \frac{|N_\rho(B)|\,|D(A) - D(B)|}{\delta^2} \leq L_C \|A - B\|_1. \tag{8}$$

Clipping preserves this Lipschitz property. To construct a self-mapping neighbourhood, let $z(t)$ be the non-negative transport evolution of A_0 and the fixed births with reaction omitted. Strong continuity gives $D(z(t)) \geq 3D_0/4$ on a sufficiently short interval. Choose $0 < R < D_0/(4\|w\|_\infty)$ and the closed tube of non-negative paths $A(0) = A_0$ with $\|A - z\|_{C([0,T];X)} \leq R$. Every path in this tube has $D(A(t)) \geq D_0/2$ and uniformly bounded X norm. Define $\mathcal{T}A$ by propagating the linear cohort system with feedback evaluated from A, while keeping births and interface maps fixed. Positive propagation preserves the cone. Bounded coefficients give $\|\mathcal{T}A - z\|_{C([0,T];X)} \leq C_0 T$; variation of constants and Gronwall's inequality give

$$\|\mathcal{T}A - \mathcal{T}B\|_{C([0,T];X)} \leq C_1 T e^{C_1 T} \|A - B\|_{C([0,T];X)}, \tag{9}$$

where the constants are uniform on this tube. Choosing T so that $C_0T \le R$ and $C_1Te^{C_1T} < 1$ makes the map a contraction of the complete tube into itself. Its fixed point is the unique local non-negative mild solution. Continuation applies while the population norm, coefficient bounds and a positive denominator bound persist. The implementation assigns zero index to an empty weighted population; that convention lies outside the positive-denominator argument.

2.3 Numerical implementation and verification design

The established application uses 60 one-month age bins, a 60-month unvaccinated warm-up and 120 recorded months. Each recorded step inserts newborn survivors, applies dose opportunities in descending dose order, stores pre-reaction states, freezes the current feedback, integrates disease and mortality reactions, and shifts survivors by one age bin. Recorded-window sequela stocks start at zero after the warm-up. Descending order prevents two administrations to the same child during one opportunity step. Exact per-cell reaction propagation is available when coefficients are frozen:

$$A^+_{gj} = A^-_{gj}e^{-\mu_{gj}h}, \qquad E_{e,gj} = A^-_{gj}\lambda_{e,gj}\frac{1-e^{-\mu_{gj}h}}{\mu_{gj}}, \tag{10}$$

with the continuous limit $E = A^-\lambda h$ at $\mu = 0$. The historical solver uses RK45; the refinement bridge uses Eq. 10. Their one-month outputs are checked under the same inputs. This algebraic bridge isolates reaction integration from the remaining transport and timing conventions.

We refine with $h = \Delta a = 1/k$ months, $k = 1, 2, 4, 8, 16$, preserving each dose opportunity at its integer attained age. A monthly conditional probability is applied once on crossing that age. Monthly birth counts are subdivided into k equal inflows, corresponding to an explicit within-month uniform-birth interpolation. Age-dependent coefficients retain their piecewise-constant one-month definition; annual target histories remain fixed. Feedback is updated each substep. Thus the bridge studies the continuous-time interpretation of births, transport and feedback, while its $k = 1$ member reproduces the declared monthly scheme. Its differences reflect these timing conventions as well as reaction integration.

The finest computed grid supplies a finite numerical reference for QALY gains. We plot gains across grid sizes and retain the complete numerical ledger in the repository source data. Annual economic timing conventions remain fixed. A separate transport test with zero initial population, $b(t) = 100[1 + 0.2\sin(t)]$, constant death hazard 0.03 per month and no vaccine has a closed characteristic solution. Cell-mass and integrated-population errors are evaluated at eight months for $k = 1, 2, 4, 8, 16, 32$. Its observed order characterises this linear birth–transport test; convergence of the nonlinear application requires separate analysis.

A controlled five-state continuous-time chain is evaluated by matrix exponentiation and numerical for-

ward integration at four times. At each time, 1,000 independent multinomial samples of 100,000 individuals check terminal occupancies using base seed 20261004. Separately, the retained closed-cohort count microsimulation contains 1,000 independently seeded replicates in each of four configurations, using two otitis-media conventions and vaccination/no-vaccination. This revision rechecks their stored summaries and discrete monthly event kernel. Its Monte Carlo intervals quantify uncertainty in the simulated means.

The constructed verification generator has $Q_{01} = 0.20$, $Q_{12} = 0.15$, $Q_{23} = 0.08$ and $(Q_{04}, Q_{14}, Q_{24}, Q_{34}) = (0.010, 0.012, 0.018, 0.022)$ per month. All other off-diagonal entries are zero, $Q_{ii} = -\sum_{j \neq i} Q_{ij}$, and state 4 is absorbing. The initial probability is $p_0 = (1, 0, 0, 0, 0)^\top$. The observation times are $t = 0.25, 1, 3, 12$ months; each terminal sample uses seed $20261004 + \lfloor 100t \rfloor$, with $N = 100{,}000$ individuals and 1,000 replicates. This chain is a fixed-rate verification example distinct from the PCV dose-state application.

For the constructed continuous-time chain, the terminal counts for N independent individuals follow a multinomial distribution with probabilities $p(t)$. Thus

$$\mathbb{E}[N_i(t)] = Np_i(t), \qquad \mathrm{Var}[N_i(t)] = Np_i(t)\{1 - p_i(t)\}, \qquad \mathrm{Cov}[N_i(t), N_j(t)] = -Np_i(t)p_j(t) \quad (i \neq j). \tag{11}$$

We reconstruct the seeded terminal samples and compare their count means and standard deviations with these expressions. Since state 4 is absorbing and initially empty, $p_4(t)$ is also the model-implied probability of absorption by time t. Central 95% binomial prediction intervals describe terminal-count variation conditional on the fixed generator and N. Forward ODEs provide these probabilities; the stochastic representation specifies the associated count distribution and event interpretation. The four observation times use independent terminal draws. In the retained discrete cohort summaries, reported deaths are conditional fatal-event rewards; actual integer fatality distributions would require separately saved event counts.

2.4 PCV application: population, schedules and evidence status

The pneumococcal conjugate vaccine (PCV) application uses five alive states: non-programme, enrolled/no dose, one dose, two doses and booster. An illustrative annual birth input of 1,384,610 and 90% newborn enrolment are held fixed. A seasonal monthly birth profile is an assumption; uniform births are also tested. WHO guidance recommends infant PCV as 2p+1 or 3p+0, beginning from six weeks of age. For 2p+1, the primary-dose interval is normally at least eight weeks, with a booster at 9–18 months; specified programme reasons can justify a four-week interval. Interrupted series resume without restarting [12,13]. The 2-, 4- and 12-month strategy is an illustrative 2p+1 programme rule. National schedules require local epidemiological and programme decisions. A conditional 1p+1 option for established population immunity is outside the strategies evaluated here. The surviving newborn boundary is $b(t) = B(t)(1 - q_N)(1 - c, c, 0, 0, 0)^\top$, with

gross birth inflow $B(t)$, enrolment $c = 0.90$ and neonatal probability $q_N = 0.0103$. PCV10-GSK, PCV13 and PCV10-SII are evaluated separately with a shared solver.

Dose-1 relative direct protection is 0.45, two-dose protection 1.00 and booster protection 1.10, with endpoint-specific absolute effectiveness capped at 0.95. Two-dose relative protection becomes 0.85 at age 24 months and 0.70 at 36 months; booster protection becomes 0.95, 0.80 and 0.70 at ages 24, 36 and 48 months. These are attained-age structural assumptions. Time since the actual administration is not tracked. Carriage weights do not wane in the current implementation.

The delay configuration applies conditional opportunities at ages 2–5, 4–8 and 12–16 months. Finite opportunity windows generate both lateness and incompletion. Six deterministic schedules, a delayed/incomplete strategy, one-dose-only delivery and hypothetical completion recovery are compared with waning and gradual build-up held enabled. Recovery adds 0.20 conditional opportunities for already enrolled children at ages 6–11, 9–17 and 17–23 months. Its eligible population excludes older non-programme children; implementing a clinical recovery policy would additionally require every minimum dose interval. All numerical maps are given in the repository parameter records.

Input evidence is classified as directly verified, transferred, computationally derived or unverified illustrative. Epidemiological studies and assessments motivate incidence and sequela assumptions [14–16]; product immunogenicity motivates sensitivity to bridging [17]. The SII direct profile equals the PCV13 profile multiplied by the retained summary multiplier 0.8203164. Its original row-level extraction is unavailable, and the visible serotype weights sum to 109%. The multiplier is therefore a frozen scenario coefficient with reproducible endpoint multiplication and unresolved empirical identification. The source ledger records this distinction, together with the disclosed administrative and caregiver-cost constants whose original records remain unavailable.

Table 1: Key computational inputs and their evidence status. Acute endpoint vectors are ordered as inpatient pneumonia, outpatient pneumonia, meningitis, sepsis and AOM. IPD/PN/AOM profile vectors refer to invasive pneumococcal disease, pneumonia and AOM. The repository contains the complete parameter and transformation ledger.

Parameter	Retained value and unit	Evidence status
Annual births	1,384,610 births/year	Illustrative demographic input
Population and horizon	Ages 0–59 months; 60-month unvaccinated warm-up; 120 recorded months	Declared model design
Enrolment and mortality	Newborn enrolment $c = 0.90$; neonatal $q_N = 0.0103$; under-five $q_5 = 0.020$	Frozen probabilities; mortality allocation disclosed in Methods
Incidence inputs	Pneumonia 1,771; meningitis 12; sepsis 124; all-cause AOM 9,570, per 100,000 child-years	Transferred/inherited; exact national values unverified

Parameter	Retained value and unit	Evidence status
Disease-age masses	Pneumonia/sepsis: $(0.50, 0.25, 0.15, 0.10)$ and meningitis: $(0.65, 0.18, 0.07, 0.10)$ at 0–11/12–23/24–35/36–59 months. AOM: $(0.1869, 0.1889, 0.6242)$ at 0–11/12–23/24–59 months	Frozen age-allocation scenarios
Severity and fatality	Severe-pneumonia share 0.389; meningitis/severe-pneumonia/sepsis fatality probabilities $(0.032258, 0.014241, 0.08)$	Inherited/transferred scenario inputs
AOM attribution	$\theta = 0.303$	Frozen pneumococcal-attribution proxy; primary detection denominators differ
Direct VE profiles	IPD/PN/AOM: GSK $(0.500, 0.220, 0.530)$; PCV13 $(0.602, 0.190, 0.460)$; SII 0.8203164086× PCV13	Endpoint profiles; SII summary bridge lacks original row-level identification
Dose opportunities	On time: probability 1 at 2, 4 and 12 months. Delayed: dose 1 $(.65, .57, .67, .60)$ at 2–5; dose 2 $(.70, .45, .30, .20, .10)$ at 4–8; booster $(.55, .35, .25, .15, .10)$ at 12–16 months	Conditional on remaining in the source state; finite windows permit incompletion
Direct dose/waning	Dose multipliers $(.45, 1, 1.10)$, absolute VE cap .95; two-dose protection becomes .85/.70 at 24/36 months; booster becomes .95/.80/.70 at 24/36/48 months	Attained-age structural assumptions; actual time since dose not tracked
Indirect index	$\rho = (0, 0, .15, .34, .40)$; age weights $(0, 1, .8, .35)$ at 0–5/6–17/18–23/24–59 months; $\eta = .88$; $s = 1$	Structural proxy; not fitted carriage prevalence
Annual target sequences	Programme-year indices 0–5: GSK .59/.49/.39/.32/.28/.26; PCV13 .54/.38/.32/.2864333333/.2528666667/.2193; SII .2211 at every index; final target retained thereafter	Inherited scenario histories; not calibrated programme trajectories
Sequela probabilities	Cognitive/epilepsy/hemiplegia/hearing after meningitis survival: $(.016, .022, .032, .046)$; implant/special education conditional on hearing: $(.025, .2574)$	Transferred/inherited; endpoint and denominator transfer require verification
Acute valuation	Utilities $(.9941, .9963, .9768, .9921, .995)$; durations $(8.74, 7, 18.41, 12.44, 7)$ days	Transferred state utilities and unverified durations; source observation-window conversion unresolved
Chronic utilities	Cognitive/hearing/epilepsy/hemiplegia: $(.62, .91, .83, .3903)$	Transferred valuations; hemiplegia uses severe-brain-damage proxy
Acute medical costs	$(214.6505, 15.6214, 557.7670, 572.4272, 12.9223)$ scenario USD/event	Unverified illustrative local aggregates; complete societal cost ledger in repository
Programme unit costs	GSK/SII procurement $2/1.03 = 1.94174757$; PCV13 4.22; delivery $2.10/1.03 = 2.03883495$, scenario USD/dose. Procurement multiplied by 1.05	Price scenarios and 3% alignment proxy; common real price year unresolved
Mortality valuation	Annual discount .03; assumed life expectancy 74 years; mean death age 2.5 years	Fixed valuation approximation with recorded-window morbidity

2.5 Indirect-protection functional and endpoint accounting

For endpoint e and product v, let IRR denote an incidence rate ratio and VE the direct vaccine effectiveness. Define an annual target-derived multiplier

$$H^*_{v,e}(y) = \mathrm{clip}_{[0,1]}\left[\frac{\mathrm{IRR}_v(y)}{1-\mathrm{clip}_{[0,0.95]}(c\eta VE_{v,e})}\right], \qquad c = 0.90, \quad \eta = 0.88. \tag{12}$$

The inherited annual scenario targets use $\eta = 0.88$ as a fixed reference normalisation, distinct from receipt generated by the dose maps. Applying one target history across distinct endpoints is a structural assumption. Target histories and clipping make the implied saturation reproducible.

The state weights are $\rho = (0, 0, 0.15, 0.34, 0.40)$, and age weights are $w(a) = 0, 1, 0.8, 0.35$ for ages 0–5, 6–17, 18–23 and 24–59 months. Define

$$C[A](t) = \frac{\sum_g \rho_g \int_0^L w(a)A_g(t,a)\,da}{\sum_g \int_0^L w(a)A_g(t,a)\,da}, \tag{13}$$

$$r[A](t) = \mathrm{clip}_{[0,1]}\left(\frac{C[A](t)}{c\eta\rho_{\mathrm{D2}}}\right), \qquad \rho_{\mathrm{D2}} = 0.34, \tag{14}$$

$$H_{v,e}[A](t) = 1 - s\{1 - H^*_{v,e}(y(t))\}r[A](t), \qquad 0 \le s \le 1. \tag{15}$$

Here $\rho_{\mathrm{D2}} = 0.34$ is the reference two-dose coefficient. Zero weighted population gives $C = r = 0$. C is a structural dose-state proxy. Local carriage studies motivate the age and schedule structure [5, 18]; the weights and target histories remain coefficients requiring calibration. The distinct quantities C, r and H specify the proxy, its normalisation and the endpoint multiplier.

For completed age month $a \in \{0, \ldots, 59\}$, let $w_e(a)$ be a disease-specific monthly allocation with $\sum_a w_e(a) = 1$. Within each declared age band, its total mass is distributed uniformly across months. A retained incidence input i_e per 100,000 child-years is mapped to

$$p^0_e(a) = 1 - \exp\{-5i_e w_e(a)/10^5\}. \tag{16}$$

Thus $5i_e/10^5$ is integrated five-year exposure for an ideal survivor under the allocation; i_e is not a monthly probability. Table 1 specifies the retained band masses. For AOM, this expression defines the all-cause baseline and the attribution factor θ is applied as described below.

For a baseline monthly endpoint probability $p^0_e(a)$, direct dose-state effectiveness $VE_{v,e,g}(a)$ and current

multiplier H, the implementation uses

$$p_{v,e,g} = \mathrm{clip}_{[0,1]}\{p_e^0(a)[1 - VE_{v,e,g}(a)]H_{v,e}[A](t)\},$$
$$\lambda_{v,e,g} = -\log[1 - \min(p_{v,e,g}, 1 - \varepsilon)], \qquad \varepsilon = 10^{-12}. \tag{17}$$

The final cap is the numerical guard in the code; current scenario probabilities remain below it. Rates have inverse-month units. Invasive pneumococcal disease (IPD) comprises the meningitis and sepsis endpoint group used for direct protection. Scaling the probability before transformation is generally different from multiplying a baseline hazard. Disease episodes accumulate as recurrent rewards, and fatal-event hazards remove alive mass.

For an endpoint probability p and retained fatality coefficient f, the inherited translation is

$$\lambda = -\log(1 - p), \qquad \mu_{\mathrm{fatal}} = -\log(1 - pf), \qquad f_{\mathrm{eff}} = \frac{\mu_{\mathrm{fatal}}}{\lambda} \quad (\lambda > 0). \tag{18}$$

Here f is the endpoint CFR for meningitis or sepsis, and the severe-pneumonia fraction times its CFR for the pneumonia endpoint. For $0 < p < 1$ and $0 < f < 1$, $f_{\mathrm{eff}} < f$, with $f_{\mathrm{eff}} \to f$ as $p \to 0$. Since both rewards integrate the same alive exposure, f_{eff} is the fatal-event-to-episode ratio for a fixed cell. Meningitis sequela sources retain $(1 - f_{\mathrm{men}})E_{\mathrm{men}}$ as a separate coefficient convention; subtracting the model's meningitis fatal-event reward would instead give $(1 - f_{\mathrm{eff,men}})E_{\mathrm{men}}$. The application preserves these inherited coefficients, making this a reward-accounting approximation to a joint episode–fatality process. Across all 144 retained trajectories, applying the nominal CFRs to episode rewards increased deaths by 0.014–0.081% relative to the inherited fatal-event rewards, holding the recorded alive exposure fixed. This is a reward-translation audit on stored trajectories; the complete diagnostic and reconstruction checks accompany the code archive.

Acute otitis media (AOM) is counted as a pneumococcal-attributable component using $p^0_{\mathrm{AOM}} = \theta p^0_{\text{all-cause AOM}}$ before hazard transformation. The frozen $\theta = 0.303$ is an attribution scenario proxy. A Vietnamese hospital study reported 30.1% culture detection and 41.1% combined culture/PCR detection in selected suppurative AOM cases [19]; these detection proportions do not identify the national causal fraction. We test $\theta = 0.301$, 0.411, 0 and 1 as detection-related and structural alternatives. They are scenario settings rather than confidence limits. The value 1 applies effects to the entire AOM baseline as a mathematical bound; historical all-cause accounting instead retains the unaffected component.

To avoid overlapping mortality inputs, set neonatal death probability $q_N = 0.0103$ and birth-to-60-month

all-cause death probability $q_5 = 0.020$. A constant residual post-neonatal hazard is

$$\mu_* = -\frac{1}{60}\log\left(\frac{1-q_5}{1-q_N}\right), \qquad \mu_{\mathrm{bg}}(a) = \mu_* - \mu^0_{\mathrm{dis}}(a), \qquad \mu_{v,g}(t,a) = \mu_{\mathrm{bg}}(a) + \mu_{v,g,\mathrm{dis}}(t,a). \tag{19}$$

Here μ^0_{dis} is the unvaccinated disease hazard under the frozen baseline inputs. All age-bin background hazards are non-negative. Neonatal mortality is applied as a boundary filter, followed by 60 model months of unvaccinated survival, giving exactly $1 - q_5$ under this allocation. The background remains fixed when CFRs or disease burden are perturbed. This constant-hazard accounting convention is compared with the inherited construction and a zero-background scenario; an empirical age-specific cause-deleted life table would replace the boundary timing and residual allocation.

2.6 Output functionals, factorial attribution and sensitivity

For endpoint intensity λ_e, event totals are $\int\int \sum_g \lambda_{e,g} A_g \, da \, dt$. Reported deaths count pneumococcal fatal-event rewards; neonatal and background hazards determine exposure and population balance. The economic module converts events and sequela stocks into discounted costs and QALY losses. Acute losses use endpoint durations and disutilities; mortality losses use average death age 2.5 years and remaining-life valuation followed by calendar-event discounting. Chronic sequela losses and recurring costs are integrated over the ten-year recorded window using annual stocks; one-time sequela costs use actual incident survivor flows. Existing sequela stocks use aggregate alive survival and are not transported by age. The resulting ledger combines remaining-life mortality valuation with recorded-window morbidity accounting and is an illustrative output functional rather than a complete lifetime cost-effectiveness analysis. The annual discount rate is 3%; cost-year alignment is a retained scenario conversion, with exceptions identified in the source ledger. These are declared approximations. Utilities are transferred across health-state descriptions; the hemiplegia value 0.3903 uses a severe-brain-damage valuation [20]. Acute duration and transferred utility are separate scenario inputs: the source elicited preferences for outcomes with specified disease courses, whose conversion to per-time health-state weights remains unverified. Other utility and delivery studies provide valuation context [21–24]. Unavailable administrative and caregiver-cost aggregates remain scenario constants. A common real price year has not been established; reported monetary values are scenario USD with explicit proxy transformations and source-year exceptions.

For $E_{e,y}$ acute events, duration d_e in days and utility u_e, the undiscounted acute loss is $E_{e,y} d_e (1-u_e)/365$. Future quantities use annual discount $d = 0.03$: disease and sequela quantities in year y receive $(1+d)^{-(y-1/2)}$, and dose costs receive $(1+d)^{-(y-1+0.35)}$. For assumed death age $a_d = 2.5$ years and remaining lifetime

$R = 74 - a_d$, set $n = \lfloor R \rfloor$ and $\tau = R - n$. The event-relative remaining-life weight is

$$W_Q(a_d) = \sum_{k=0}^{n-1} (1+d)^{-(k+1/2)} + \tau(1+d)^{-(n+\tau/2)} = 29.74234703. \tag{20}$$

This weight is then discounted from the calendar event. For stored birth-discounted annual productivity contributions q_k, the corresponding fatal-event weight is $W_P(a_d) = (1+d)^{a_d} \sum_{k>\lfloor a_d \rfloor} q_k / 1.03 = 44{,}314.83832662$ scenario USD. The factor $(1+d)^{a_d}$ converts birth-discounted contributions to the event-relative scale. Separately, an input labelled source year $s > 2023$ is aligned by $c_s/(1.03)^{s-2023}$; unlabelled inputs and the retained PCV13 procurement price are unchanged. This price-alignment proxy is distinct from outcome discounting. The repository preserves all category-specific costs, productivity contributions and source-year exceptions.

For strategy z, let L_z be loss of quality-adjusted life years (QALYs) and K_z total cost. Gains are $Q_z = L_{\mathrm{NV}} - L_z$, incremental costs $\Delta K_z = K_z - K_{\mathrm{NV}}$, and ICER $\Delta K_z / Q_z$ when $Q_z \neq 0$. Outputs are standardised per 100,000 births during the recorded window, including benefits to children present at entry. This denominator is not the total number of exposed children. Negative ICERs are accompanied by dominance status and are not used to rank products.

The full 2^3 factorial independently enables delayed/incomplete dose states (D), attained-age direct waning (W) and an additional state-dependent herd ramp (G). Reference 000 uses on-schedule dose maps, no waning and $H = H^*(y)$; 111 enables all three mechanisms. $G = 0$ retains the annual target history and does not switch herd off. Matched herd-off runs set $H = 1$ for each D/W configuration. For any output Y, the interaction for subset S is

$$\beta_S = \sum_{T \subseteq S} (-1)^{|S|-|T|} Y(\mathbf{1}_T), \qquad \phi_i = \sum_{S \ni i} \frac{\beta_S}{|S|}. \tag{21}$$

The Shapley allocations sum to $Y(111) - Y(000)$ and distribute modelled interactions across the three mechanisms. A separate scheduling comparison fixes W/G enabled, so its on-time reference is 011.

Indirect QALY contribution is reproducibly defined by

$$Q_{\mathrm{indirect}} = L_{\mathrm{off}} - L_{\mathrm{on}} = Q_{\mathrm{on}} - Q_{\mathrm{off}}, \qquad f_{\mathrm{indirect}} = \frac{Q_{\mathrm{indirect}}}{L_{\mathrm{NV}} - L_{\mathrm{on}}}. \tag{22}$$

The paired runs hold all inputs and dose rules fixed while allowing downstream survival and stocks to respond. The contribution includes benefits among vaccinated and unvaccinated children. Costs are treated analogously; ICER contrasts describe changes in the resulting cost-to-gain ratios.

The application comprises 37 core, 27 schedule and 80 perturbation/comparator configurations, including

deliberately repeated bridge cases. Price, direct effectiveness, CFR, sequela costs, utility loss and disease burden are varied by factors 0.75 and 1.25. Additional settings vary waning, herd strength, dose completion, birth seasonality and costing perspective. Direct-VE perturbations fix baseline H^* to prevent algebraic compensation; state-dependent feedback remains responsive. No joint distributions or correlations are assigned, and these analyses do not constitute probabilistic sensitivity analysis (PSA). New verification, refinement, AOM and mortality experiments are reported separately from these 144 application configurations.

3 Results

Three findings organise the results. First, matched deterministic and stochastic representations agree on expected outcomes, while the stochastic formulation also characterises conditional terminal variation and absorption probabilities. Second, populations with identical age and dose marginals produce different feedback and disease-event totals when their age–dose associations differ. A reduction based only on those marginals therefore needs additional information or a closure assumption. Third, retaining the structure permits separate evaluation of dosing, waning and indirect-protection build-up: their combined effect reduces PCV QALY gains by 10.2–18.4%, and the paired indirect pathway contributes 61.0–69.7% of full-scenario gains. The PDE provides an explicit representation of the joint structure and shared feedback; an ODE or Markov model retaining the same information can represent the same process. Numerical refinement and sensitivity analyses distinguish these structural findings from discretisation and input assumptions.

3.1 Matched representation and numerical verification

The controlled continuous-time forward equation matched the matrix exponential with maximum absolute state-probability error 7.6e-13. In the retained independently seeded cohort audit, 16 of 16 deterministic comparisons lay within pointwise 95% Monte Carlo intervals for the simulated means; this is a diagnostic check without a simultaneous-coverage claim. The population and cell-mass errors in the manufactured transport example decreased under refinement. Its finest successive integrated-population order was 1.003.

The stochastic interpretation supplies a distribution around a mean-count summary. At 12 months, the constructed chain gives an absorption probability of 0.14606. For 100,000 independent individuals, the expected absorbing count is 14,605.58 and its exact standard deviation is 111.68; the reconstructed Monte Carlo standard deviation is 114.06. The central 95% binomial prediction interval is 14,387–14,825 individuals. These quantities describe fixed-input terminal variation in the verification chain. The repository reports all observation times and distinguishes prediction intervals from uncertainty in a simulated mean.

The exact-reaction one-month implementation reproduced the revised application's four full/NV output

ledgers within the specified bridge tolerance. At $h = 1/16$ month, full-scenario QALY gains were 1,421.53, 1,516.35, 1,596.50 per 100,000 births for GSK, PCV13 and SII, respectively. Relative to this finite reference, monthly QALY-gain differences were 0.386%, 0.417%, 0.530%; the largest difference between $h = 1/8$ and $h = 1/16$ was 0.035%. These differences include the declared birth interpolation and feedback update frequency. Fig. 1C plots the actual gain at every tested grid size, including $h = 1/16$. The complete gain-refinement ledger is retained in the repository.

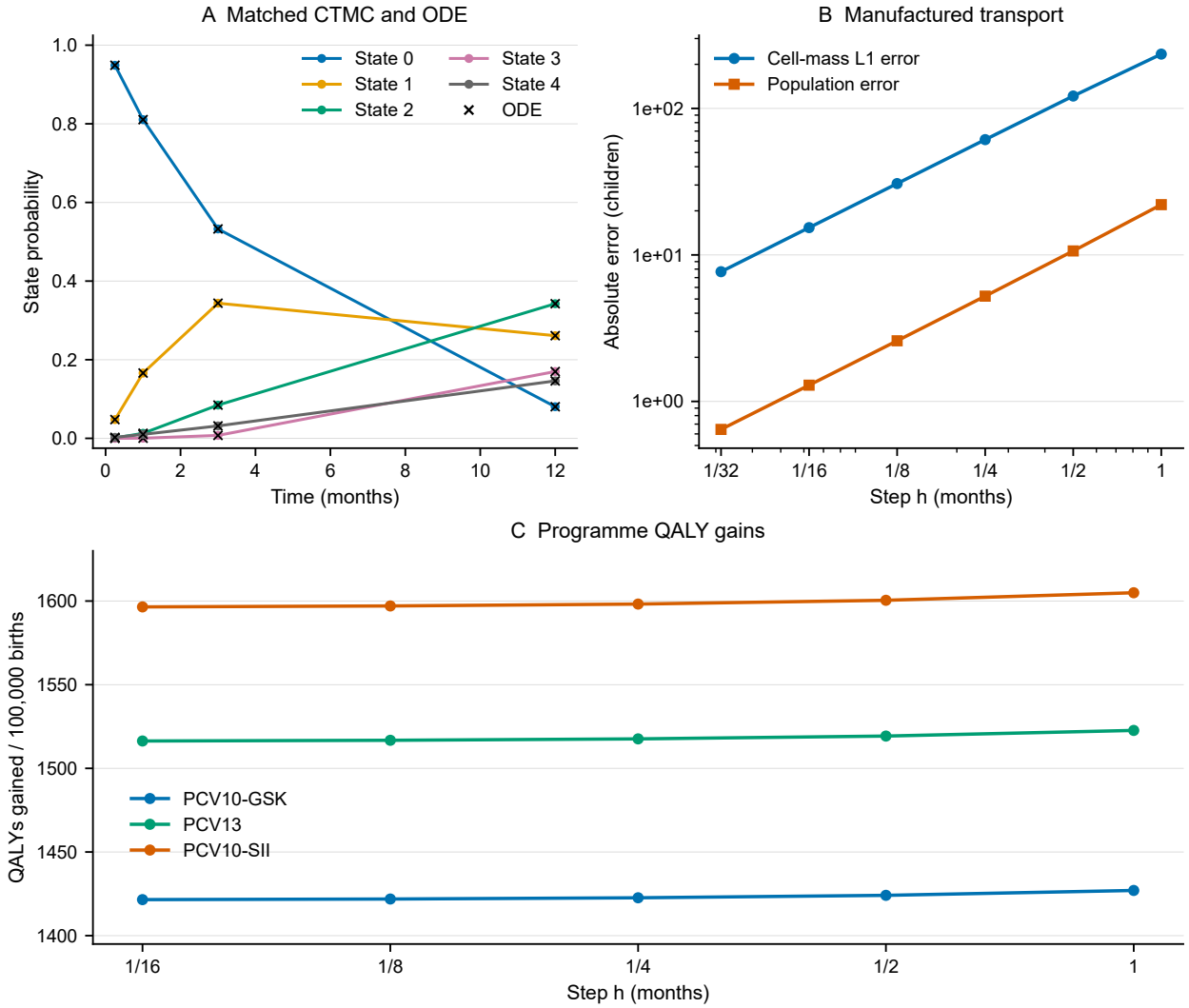


Fig 1: Expected-state verification and refinement of QALY gains. A, state probabilities in a constructed five-state continuous-time Markov chain: coloured curves are matrix-exponential solutions and black crosses are the matched forward ODE solution; their overlap verifies expected occupancies under the same generator. State 4 is absorbing; these verification states are distinct from the PCV alive dose categories. B, errors relative to an analytic manufactured transport solution, plotted against step size in months; decreasing errors test the birth/transport discretisation. C, full-scenario QALY gains per 100,000 recorded-window births for each product across all five grids, including $h = 1/16$ month as the finest finite numerical reference. Panel B measures error against an analytic solution; panel C compares application gains across grids.

3.2 Age–dose association contains information absent from both marginals

Configuration A and configuration B each contained 50,000 alive individuals, with 10,000 in each of the five programme states. Both placed 10,000 individuals in each one-month age bin beginning at 4, 8, 16, 20 and 48 months. In configuration A, the two-dose group occupied the 16-month bin and the booster group the 48-month bin; configuration B exchanged those two assignments. The other groups were unchanged: enrolled/no dose at 4 months, one dose at 8 months and non-programme at 20 months. The one-month reaction uses coefficients evaluated from each starting population and held fixed during that reaction. The weighted index changed from 0.2000 to 0.2124, and the IPD multiplier from 0.5686 to 0.5419. One-month disease events changed from 99.58 to 92.26 (-7.36%). Both initial death derivatives and event rates differed. The current observables therefore distinguish populations with identical dose and age marginals (Fig. 2).

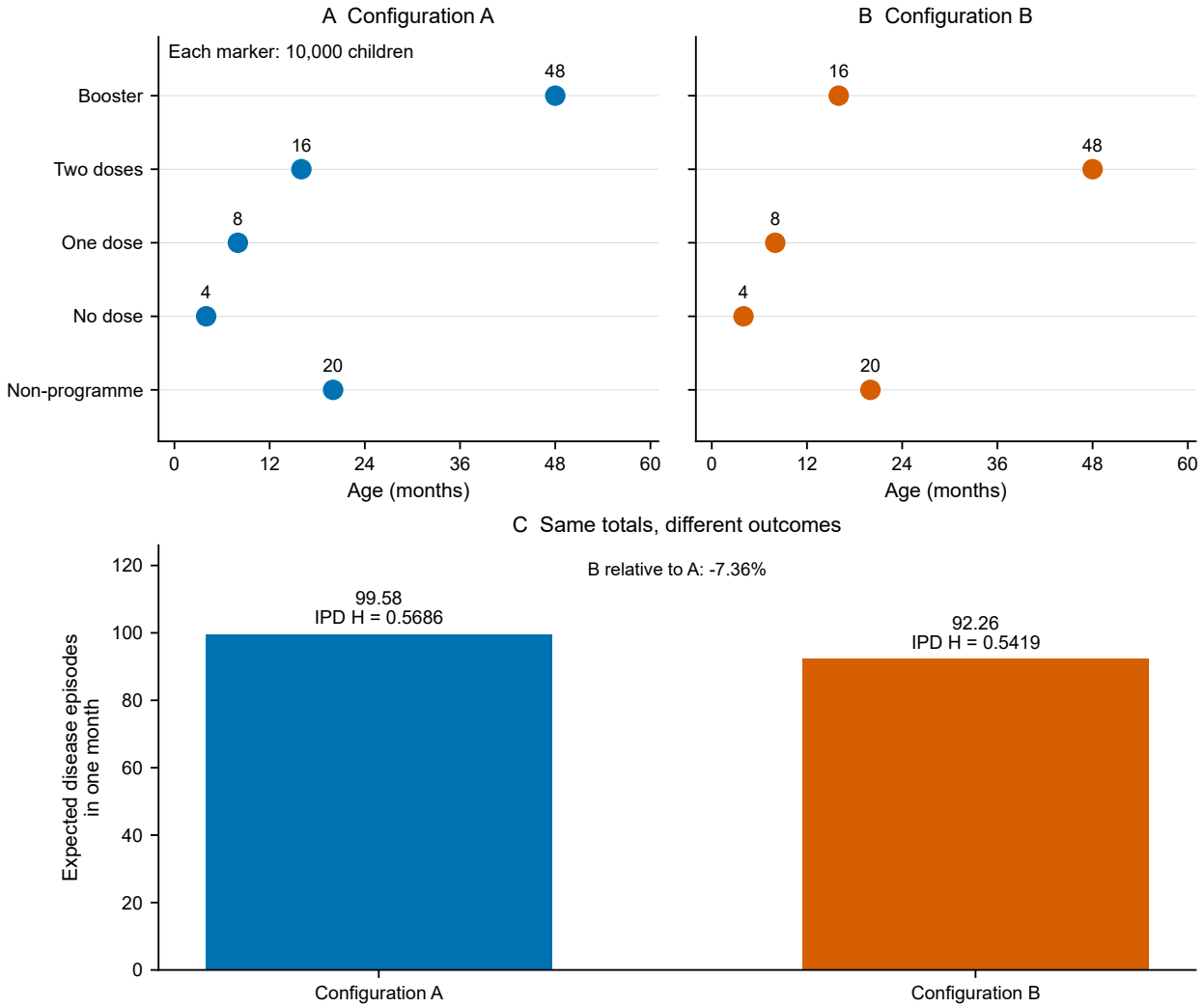


Fig 2: Joint age–dose structure distinguishes populations with identical marginals. A and B, each point represents 10,000 children in the one-month age bin beginning at the labelled age and in the indicated programme state. Configuration A places the two-dose group at 16 months and the booster group at 48 months; configuration B swaps those two assignments. Both age and dose-state totals remain identical. C, the corresponding expected disease episodes during one frozen-coefficient month are 99.58 and 92.26; annotations report the dimensionless initial IPD multipliers, 0.5686 and 0.5419. The 7.36% reduction reflects the changed joint association. These are constructed diagnostic populations.

3.3 Population trajectories expose scheduling and feedback

In the application, non-programme children from the warm-up coexist with newly enrolled cohorts. Dose-state density follows cohort diagonals and accumulates across successive births. Delayed/incomplete dosing shifts protection towards later ages and leaves some enrolled children in incomplete states. Fig. 3 presents no vaccination, on-time 011 and delayed 111 populations alongside age-integrated state totals, C, r, H and annual disease events. These trajectories connect population structure and the declared feedback to

355 aggregate gains. The repository provides product-specific trajectory panels and year-10 outcomes.

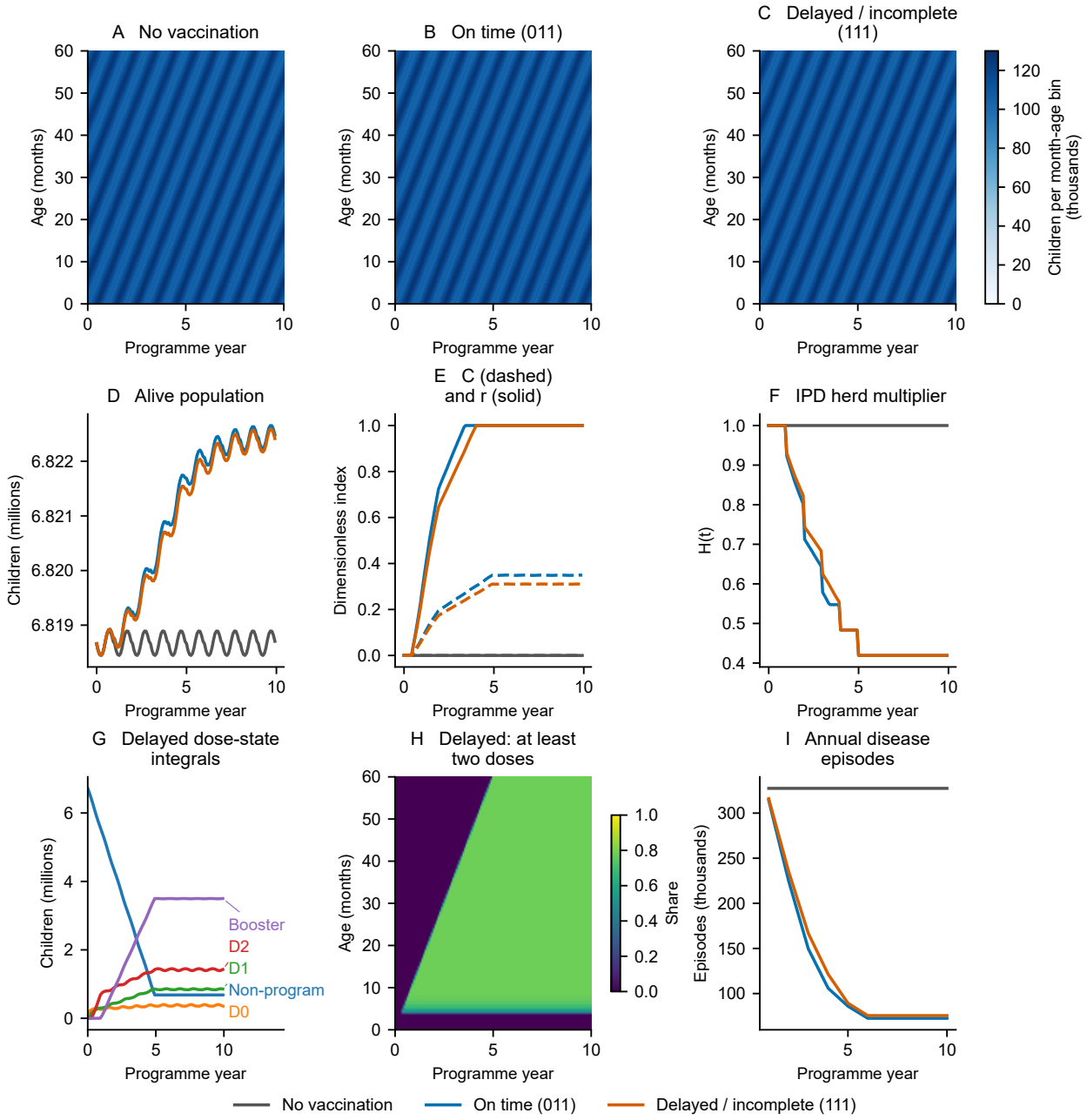


Fig 3: Age–calendar densities and integrated PCV13 trajectories explain the programme mechanism. A–C, total alive density by age and programme year under no vaccination, on-time 011 and delayed/incomplete 111; colour denotes thousands of children per month-age bin. D, age-integrated alive population under the same three strategies. E, the weighted dose proxy C (dashed) and normalised build-up index r (solid); F, the corresponding IPD multiplier H, where smaller values mean stronger modelled indirect protection. G, age-integrated dose-state populations under 111. H, the age-specific share with at least two doses under 111. I, annual disease episodes. State integrals and indices use the declared pre-reaction timing. The paired scenarios retain W/G enabled, so their contrast concerns dose delivery rather than an additional switch in waning or herd build-up.

3.4 Factorial effects and paired indirect contribution

The established monthly implementation gave full-scenario gains of 1,427.0, 1,522.7, 1,605.0 QALYs per 100,000 births. Joint decrements relative to 000 were 10.2%, 11.1%, 18.4%. This contrast changes D, W and G together. The gradual-build-up factor accounted for 64.6%, 67.5%, 86.1% of the joint QALY decrement under Shapley allocation. Matched full-scenario herd comparisons yielded indirect shares of 62.3%, 61.0%, 69.7% of total gains. The indirect share measures switching off the whole indirect pathway, whereas the ramp decrement measures the timing of its build-up. Table 2 gives the endpoint and economic contrasts; Fig. 4 decomposes the changes.

Table 2: Monthly application outcomes versus no vaccination per 100,000 recorded-window births. Deaths are pneumococcal fatal-event rewards. Cost differences are million scenario USD; all vaccine rows are dominant under this societal scenario.

Product	DWG	Cases averted	Deaths averted	QALY gain	Cost diff.	ICER
PCV10-GSK	000	14,609.5	62.7	1,588.6	-4.397	-2,768
PCV10-GSK	111	13,131.3	56.9	1,427.0	-3.970	-2,782
PCV13	000	16,083.8	67.5	1,713.3	-4.354	-2,542
PCV13	111	14,208.3	60.6	1,522.7	-3.873	-2,543
PCV10-SII	000	18,190.7	76.2	1,966.4	-5.729	-2,913
PCV10-SII	111	14,969.1	63.6	1,605.0	-4.605	-2,869

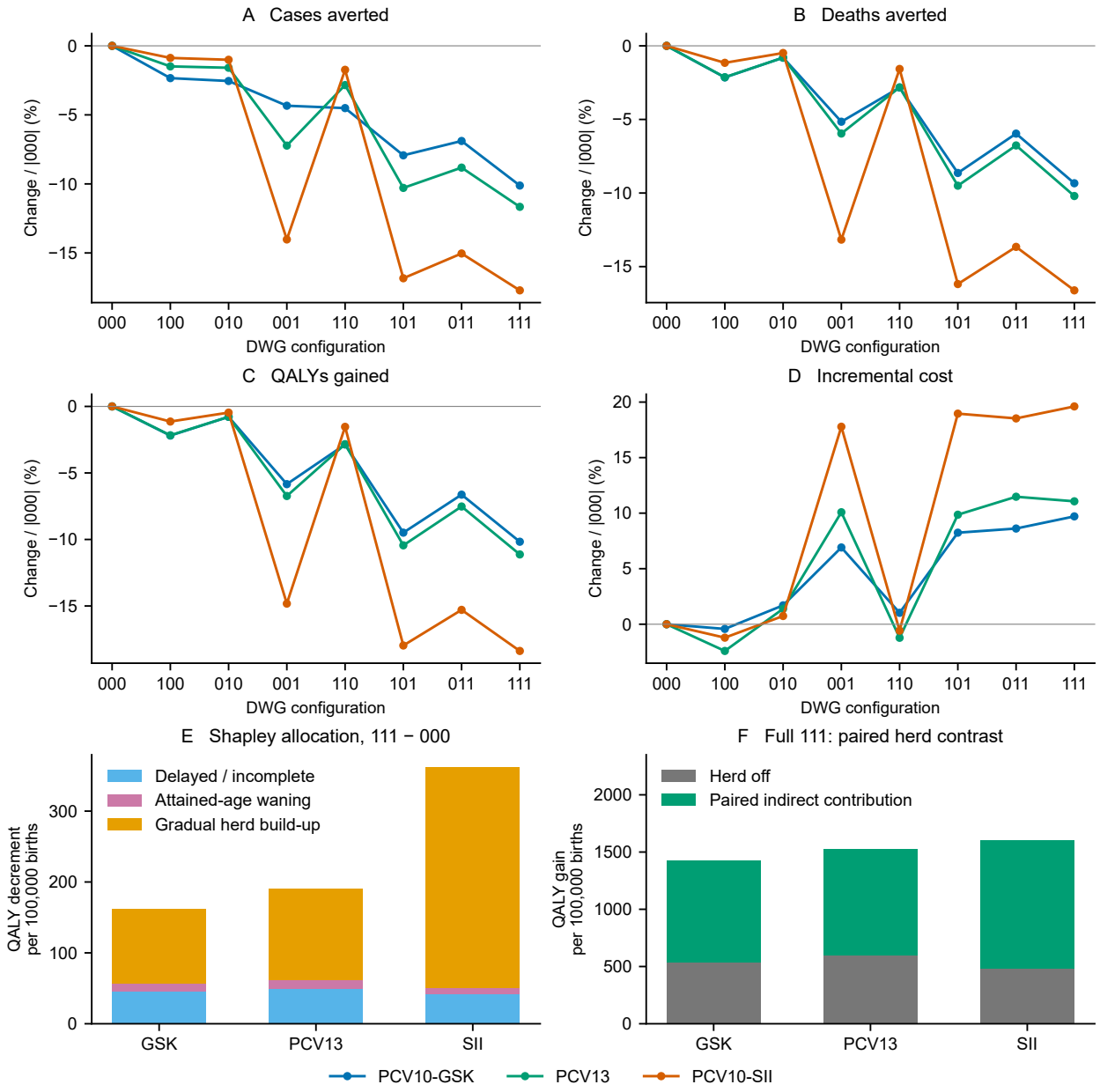


Fig 4: Separate mechanisms and paired herd contributions under a common solver. A–D, percentage changes in cases averted, pneumococcal deaths averted, QALY gains and incremental costs relative to each product's 000 reference, calculated as $100\{Y(z) - Y(000)\}/|Y(000)|$. The configurations are distinct factorial settings, not successive time points. D enables delayed/incomplete delivery, W attained-age waning and G gradual state-dependent herd build-up. A positive change in negative incremental cost means less saving. E, Shapley allocation of the total QALY decrement from 000 to 111, distributing interactions across the three mechanisms. F, full111 gains with herd disabled (grey) plus the difference from the matched herd-on run (green). G=0 retains target-derived protection; the herd-off counterfactual instead sets H=1. All gains are per 100,000 recorded-window births; product-specific assumptions preclude a clinical product ranking.

3.5 Schedule strategies and structural robustness

All schedule experiments retained W/G enabled. On-time 011 gained 1,483.3, 1,584.4, 1,665.7 QALYs per 100,000 births, compared with 1,427.0, 1,522.7, 1,605.0 under delayed/incomplete delivery. One-dose-only strategies produced substantially smaller gains, while completion recovery restored part of the loss. Fig. 5 reports the comparisons; complete numerical schedule outcomes are retained in the repository.

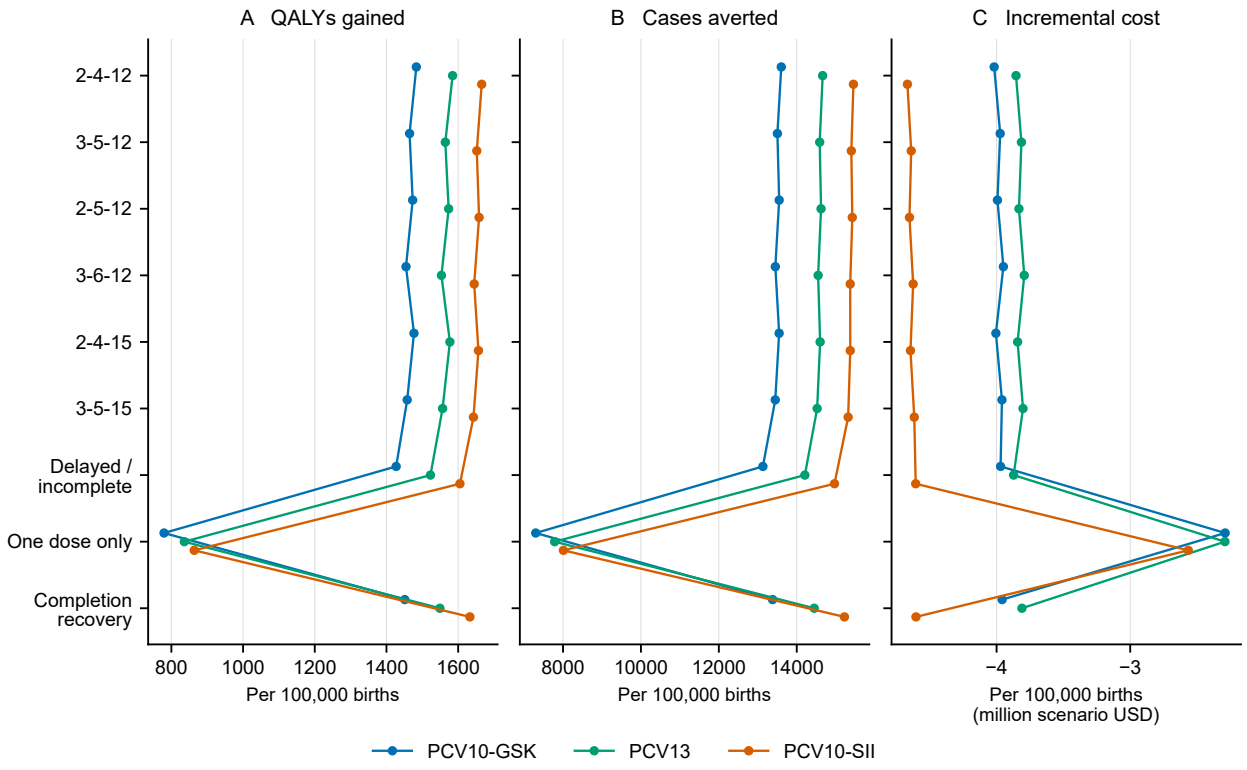


Fig 5: Expected programme outcomes under alternative dose-delivery rules, with waning and gradual herd build-up enabled. Rows give six planned schedules in months, delayed/incomplete delivery, one-dose-only delivery and completion recovery. A, QALYs gained; B, cases averted; C, incremental societal cost in million scenario USD, all versus matched no vaccination per 100,000 recorded-window births. Later protection generally reduces modelled gains, while recovery restores part of the delayed/incomplete deficit. Recovery opportunities apply only to enrolled children and are hypothetical rules. Points are deterministic outputs; connecting lines guide comparison across the categorical strategies.

Deterministic screening changed gains and costs across plausible-direction perturbations without assigning probability distributions. The restricted healthcare convention changed all three incremental costs to positive values: 133,429, 534,949, 48,141 USD per 100,000 births, with unchanged QALY gains. Thus societal cost savings do not imply healthcare budget savings.

For PCV13, the AOM attribution settings 0.301 and 0.411 gave 1,522.66 and 1,522.91 QALYs gained, respectively, versus 1,522.66 at the frozen proxy. Cost differences changed to -3.865 and -4.287 million USD. The operational proxy therefore matters more for counted AOM and costs than for mortality-dominated

QALY gains in this ledger.

The historical inherited neonatal/background construction implied approximately 30.1 deaths per 1,000 births before added pneumococcal hazards. PCV13 QALY gains became 1,521.92 in the neonatal-conditioned bracket and 1,525.73 with zero background. The revised cause-deleted allocation reconciles baseline unvaccinated survival to 20 deaths per 1,000 births; the alternative constructions quantify sensitivity to mortality accounting rather than identify an empirical life table. Fig. 6 and repository result tables provide all settings.

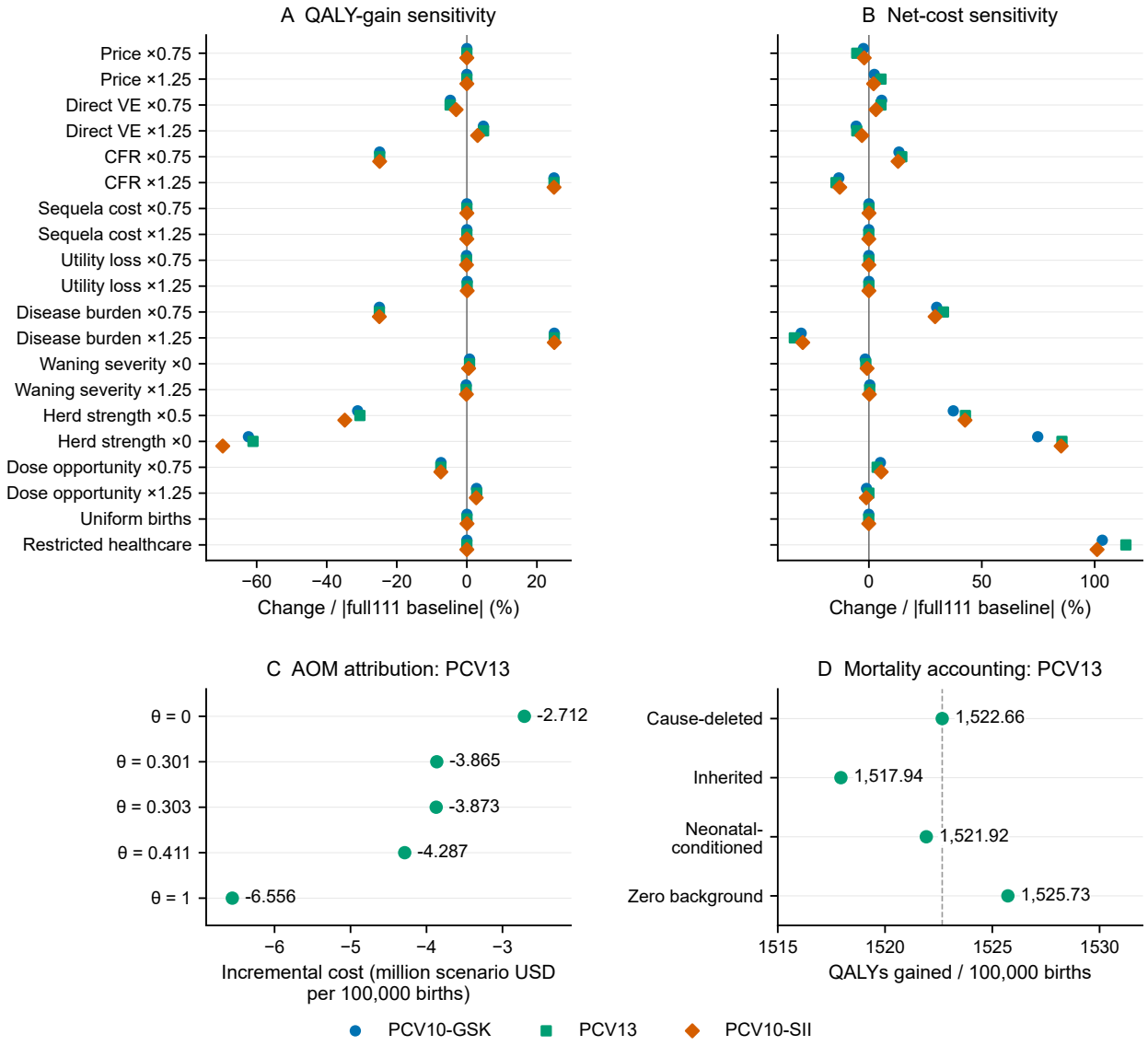


Fig 6: Horizontal comparison of deterministic perturbations and accounting scenarios. A and B, QALY-gain and incremental-cost changes for 20 retained settings and three products; the horizontal coordinate is $100(Y_{\text{perturbed}} - Y_{111})/|Y_{111}|$. Positive cost changes mean less saving or a higher net cost. Numeric setting labels give the declared multiplier or scenario value; uniform births and healthcare costs are alternative conventions. C, PCV13 incremental cost per 100,000 births under five AOM attribution settings. D, PCV13 QALY gains under four mortality conventions; the dashed line marks the revised cause-deleted convention, and the displayed axis is intentionally restricted to show small differences. Each point is a fixed-scenario result.

4 Discussion and conclusions

The study addresses three different questions that require different information: what outcomes are expected, which age–dose associations determine those outcomes, and how random outcomes vary around their expec-

tations. The ODE, PDE and Markov descriptions provide related tools for answering them. Their value lies in matching the representation to the question while keeping population boundaries, intervention rules, feedback and reward accounting explicit.

The principal methodological result is that preserving the joint age–dose distribution can matter even when both marginal distributions are known. The constructed swap changes the nonlocal index and endpoint rewards while preserving either set of totals. A reduced model based on those marginals needs a closure assumption or additional state information to distinguish the configurations. Structured transport retains that information explicitly; sufficiently detailed ODE, Markov and individual models can retain it as well.

The characteristic construction connects the structured model to familiar state-transition accounting. Under fixed external feedback, the cohort ODE and the transport characteristic have the same propagation. With endogenous feedback, cohorts interact through the shared functional. Explicit interface kernels represent scheduled dose opportunities alongside continuous reaction dynamics. Comparing representations therefore requires matched boundaries, transitions, feedback and rewards.

The ODE and Markov descriptions serve complementary analytical roles. Under the matched externally specified process, the forward ODE computes state probabilities and expected counts efficiently. The Markov law supplies their count-distribution and event interpretation; moment or backward equations can recover further distributional quantities. In the controlled example, the expected terminal count is accompanied by a count standard deviation and a conditional prediction interval. The discrete closed-cohort simulation separately checks expected rewards. Together these examples distinguish computational expectations, fixed-input count variation and Monte Carlo error in an estimated mean.

Numerical evidence has several levels. Matrix-exponential comparisons verify a controlled continuous-time chain. Seeded count simulations check matched discrete cohort expectations. Analytic reaction comparisons verify the frozen-coefficient integrator. The manufactured solution checks birth/transport behaviour, while the application refinement bridge examines timing and feedback resolution. Each test has a defined numerical target. Empirical calibration addresses a separate question: whether the input values and indirect-effect mapping describe a target programme.

The PCV application demonstrates why mechanism decomposition is useful. The matched factorial holds population input, state representation and accounting fixed while varying dosing, direct waning and indirect-protection build-up. Interactions account for differences between isolated and joint effects. Shapley allocation distributes these interactions into an additive summary conditional on the chosen target histories, age weights and dose rules. The herd-on/off comparison separately quantifies the model's response to its entire indirect pathway.

For readers evaluating vaccination programmes, the framework offers a diagnostic and an analysis de-

sign. The age–dose counterexample identifies missing joint information; characteristic comparisons specify matching conditions; and the factorial and paired-herd experiments distinguish mechanism timing from the whole indirect pathway. For policy analysts, the results identify evidence to collect: actual dose-age and completion distributions, age-specific disease risk, the evolution of carriage or indirect protection, and locally verified valuation inputs. These measurements would support calibrated comparisons of delivery and recovery strategies.

The AOM analysis connects source verification to output interpretation. The hospital detection denominators motivate alternative attribution settings, which affect counted episodes and costs more than the mortality-dominated QALY gains in this ledger. Estimating a national causal fraction would require representative attribution data. The older all-cause accounting exercise supplies a separate denominator check.

The mortality revision separates frozen background from unvaccinated disease hazards and reconciles no-vaccination survival to the stated under-five probability through a constant-hazard allocation. Empirical age-specific mortality would replace this convention. New sequela costs use incident source flows, with the retained CFR translation and sequela-source coefficients explicitly distinguished in Eq. 18. A repository migration table isolates the effects of the mortality and incident-cost accounting repairs.

Several extensions follow from the framework. Actual time-since-dose waning requires an additional elapsed-time coordinate or augmented state; the present attained-age approximation can treat a late dose as though it had aged like an early dose. Age-dependent contact, calibrated carriage, serotype replacement and adult benefits require additional transmission states and evidence. Calendar-specific enrolment and verified dose-age distributions could replace the assumed birth and completion patterns.

Economic results illustrate output functionals and show how costing perspective changes the apparent savings. The inherited ledger combines transferred utilities, annual stock approximations and proxy product assumptions; local cost verification and a common real price year remain necessary for policy use. The restricted healthcare calculation retains some sequela and special-education categories, so a payer budget-impact analysis would require a separately defined scope. Product-specific target histories and protection profiles define the three scenarios; negative ICERs carry dominance labels for each comparison with no vaccination.

The current calendar analysis uses deterministic parameter screening. An early closed-cohort report described 200 exploratory parameter draws; the original executed draws and run-specific seed remain unavailable. A reproducible joint calendar PSA requires defensible marginal distributions, correlations, and coherent treatment of direct effectiveness and the target-derived multiplier. This parameter uncertainty is distinct from individual-event variation and Monte Carlo error. The deterministic experiments estab-

lish the reported structural comparisons; calibrated policy evaluation additionally requires joint parameter uncertainty.

The repository maps the project research strands and their evidence status: exact cohort propagation and assumption tiers, ODE/count-simulation checks, age–time and open-population bridges, endpoint-specific protection, dose realism, economic accounting, factorial and schedule analysis, and verification. Historical results with inconsistent calendar discounting are identified in the development record and excluded from current effect estimates. The record includes the earlier propagation and prefix experiments, characteristic comparison and deterministic cohort screening, with distinct denominators, protection conventions and cost years reported separately.

A hybrid age–calendar formulation connects cohort state transitions to ongoing vaccination programmes, while controlled stochastic verification distinguishes expected outcomes from conditional count variation. Its contribution is the explicit preservation and interrogation of joint age–dose information, birth and intervention boundaries, and nonlocal feedback. Derivation, counterexamples and numerical checks show when aggregation loses information and how implementation assumptions alter programme comparisons.

Data and code availability

The aggregate inputs, 144 application configurations and their numerical outputs, mathematical experiments, reproduction code, revised manuscript source, publication figures and CFR-translation audit are publicly available at `https://github.com/Raymond-Zhou-2005/pcv-structured-model`, release `v1.0.1`. The fixed release is `https://github.com/Raymond-Zhou-2005/pcv-structured-model/releases/tag/v1.0.1`. It includes data definitions, parameter/source/transformation ledgers, dependencies, seed rules and SHA-256 manifests. The simulation inputs and numerical outcomes are retained from `v1.0.0`; this release incorporates the manuscript and figure revision labelled `2026-11-03-r1` and its saved-trajectory audit. Original code is licensed under MIT, and original data and text under CC BY 4.0, with third-party rights and attributions documented separately. The assets comprise aggregate scenario constants and simulated outputs. The source ledgers record published-source provenance and outstanding parameter qualifications.

Ethics

This work conducts mathematical and computational analyses without recruiting participants or accessing individual patient records.

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